

❏ Useful Wireshark Filters to Keep Handy

Purpose	Filter
TCP resets	<code>tcp.flags.reset == 1</code>
DNS errors	<code>dns.flags.rcode != 0</code>
HTTP errors	<code>http.response.code >= 400</code>
Retransmissions	<code>tcp.analysis.retransmission</code>
ARP anomalies	<code>arp.duplicate-address-detected</code>
Cleartext creds	<code>http contains "pass"</code>
Large packets	<code>frame.len > 1400</code>
ICMP only	<code>icmp</code>
Specific IP	<code>ip.addr == x.x.x.x</code>

❏ RECONNAISSANCE & SCANNING

Purpose	Filter
SYN scan (half-open)	<code>tcp.flags == 0x002</code>
SYN-ACK responses	<code>tcp.flags == 0x012</code>
TCP NULL scan	<code>tcp.flags == 0x000</code>
FIN scan	<code>tcp.flags == 0x001</code>
XMAS scan	<code>tcp.flags == 0x029</code>
RST flood / port scan indicator	<code>tcp.flags.reset == 1</code>
ICMP sweep (ping scan)	<code>icmp.type == 8</code>
ICMP replies	<code>icmp.type == 0</code>
UDP scan (ICMP unreachable response)	<code>icmp.type == 3 && icmp.code == 3</code>
OS fingerprinting (unusual TCP options)	<code>tcp.options</code>

❏ DENIAL OF SERVICE (DoS / DDoS)

Purpose	Filter
SYN flood	<code>tcp.flags.syn == 1 && tcp.flags.ack == 0</code>
ICMP flood	<code>icmp</code>
UDP flood	<code>udp</code>
Fragmentation attack	<code>ip.flags.mf == 1 ip.frag_offset > 0</code>
Smurf attack (broadcast ICMP)	<code>icmp.type == 8 && ip.dst == 255.255.255.255</code>
RST flood	<code>tcp.flags.reset == 1</code>
ACK flood	<code>tcp.flags.ack == 1 && tcp.flags.syn == 0</code>
DNS amplification	<code>dns.flags.response == 1 && udp.length > 400</code>
NTP amplification	<code>ntp.control.monlist</code>

🔍 MALWARE / C2 DETECTION

Purpose	Filter
Beaconing (periodic connections)	<code>tcp.flags.syn == 1 && ip.dst == <suspicious_ip></code>
DNS tunneling (long subdomains)	<code>dns.qry.name matches "[a-z0-9]{30,}"</code>
DNS over non-standard port	<code>dns && udp.port != 53 && tcp.port != 53</code>
HTTP C2 (unusual user agents)	<code>http.user_agent contains "curl"</code>
HTTP C2 (empty user agent)	<code>http.user_agent == ""</code>
HTTPS to unusual port	<code>ssl && tcp.port != 443</code>
IRC C2 traffic	<code>tcp.port == 6667</code>
Known RAT port (Metasploit default)	<code>tcp.port == 4444</code>
Base64 in HTTP URI	<code>http.request.uri matches "[A-Za-z0-9+/{20,}"</code>
Large DNS TXT response (exfil)	<code>dns.resp.type == 16 && dns.resp.len > 100</code>

🔍 CREDENTIAL & DATA EXFILTRATION

Purpose	Filter
HTTP Basic Auth (cleartext)	<code>http.authorization</code>
FTP credentials (cleartext)	<code>ftp.request.command == "PASS"</code>
FTP username	<code>ftp.request.command == "USER"</code>
Telnet traffic (cleartext)	<code>telnet</code>
HTTP POST with credentials	<code>http.request.method == "POST"</code>
SMTP traffic (email exfil)	<code>smtp</code>
POP3 cleartext login	<code>pop.request.command == "PASS"</code>
IMAP cleartext login	<code>imap contains "LOGIN"</code>
Large HTTP POST (exfiltration)	<code>http.request.method == "POST" && http.content_length > 5000</code>
Large outbound TCP transfer	<code>tcp.len > 1000 && ip.dst != <internal_range></code>
NTLM hashes in traffic	<code>ntlmssp</code>
Kerberos ticket requests	<code>kerberos.msg_type == 10</code>
Kerberoasting (TGS-REQ)	<code>kerberos.msg_type == 12</code>
AS-REP Roasting	<code>kerberos.msg_type == 11</code>

🔍 LATERAL MOVEMENT

Purpose	Filter
SMB traffic (internal spread)	<code>smb smb2</code>
SMB login attempts	<code>smb2.cmd == 1</code>

Purpose	Filter
Pass-the-Hash via SMB	smb2 && ntlmssp.auth
RDP traffic	tcp.port == 3389
RDP brute force	tcp.port == 3389 && tcp.flags.syn == 1
WMI remote execution	tcp.port == 135
PsExec indicators	smb2 && smb2.filename contains "PSEXESVC"
WinRM (remote PowerShell)	tcp.port == 5985 tcp.port == 5986
SSH lateral movement	tcp.port == 22
VNC remote access	tcp.port == 5900

❑ DNS ANALYSIS

Purpose	Filter
All DNS traffic	dns
DNS queries only	dns.flags.response == 0
DNS responses only	dns.flags.response == 1
Failed DNS lookups (NXDOMAIN)	dns.flags.rcode == 3
DNS server errors	dns.flags.rcode == 2
DNS ANY queries (amplification)	dns.qry.type == 255
DNS zone transfer (AXFR)	dns.qry.type == 252
DNS PTR (reverse lookup)	dns.qry.type == 12
High-entropy domain names	dns.qry.name matches "[a-z0-9]{15,}\.\.(com net org)"
DNS over TCP (large payload)	dns && tcp

❑ TLS / SSL ANALYSIS

Purpose	Filter
All TLS traffic	tls
TLS Client Hello	tls.handshake.type == 1
TLS Server Hello	tls.handshake.type == 2
TLS certificates	tls.handshake.type == 11
SSL version 2 (insecure)	ssl.record.version == 0x0002
SSL version 3 (insecure)	ssl.record.version == 0x0300
TLS 1.0 (weak)	tls.record.version == 0x0301
TLS 1.1 (weak)	tls.record.version == 0x0302
TLS 1.2 (acceptable)	tls.record.version == 0x0303
TLS 1.3 (preferred)	tls.record.version == 0x0304
Self-signed cert detection	tls.handshake.certificate
JA3 fingerprint (client)	tls.handshake.ja3

Purpose	Filter
JA3S fingerprint (server)	<code>tls.handshake.ja3s</code>

❑ ARP POISONING / MITM

Purpose	Filter
All ARP traffic	<code>arp</code>
ARP requests only	<code>arp.opcode == 1</code>
ARP replies only	<code>arp.opcode == 2</code>
Gratuitous ARP (poisoning)	<code>arp.isSolicited == 0</code>
Duplicate IP detection	<code>arp.duplicate-address-detected</code>
ARP from unexpected MAC	<code>arp.src.hw_mac != <expected_mac></code>
ARP flood	<code>arp && eth.src == <attacker_mac></code>

❑ HTTP / WEB ATTACKS

Purpose	Filter
All HTTP traffic	<code>http</code>
HTTP GET requests	<code>http.request.method == "GET"</code>
HTTP POST requests	<code>http.request.method == "POST"</code>
HTTP 200 OK	<code>http.response.code == 200</code>
HTTP 403 Forbidden	<code>http.response.code == 403</code>
HTTP 404 Not Found	<code>http.response.code == 404</code>
HTTP 500 Server Error	<code>http.response.code == 500</code>
SQL injection in URI	<code>http.request.uri contains "' OR"</code>
Directory traversal	<code>http.request.uri contains "../"</code>
Command injection	<code>http.request.uri contains "cmd="</code>
XSS attempt	<code>http.request.uri contains "<script>"</code>
Web shell access	<code>http.request.uri contains ".php?cmd"</code>
Suspicious file download	<code>http.request.uri contains ".exe"</code>
User-Agent spoofing (blank)	<code>http.user_agent == ""</code>
Nikto web scanner	<code>http.user_agent contains "Nikto"</code>
SQLmap scanner	<code>http.user_agent contains "sqlmap"</code>

❑ TCP PERFORMANCE & ANALYSIS

Purpose	Filter
All retransmissions	<code>tcp.analysis.retransmission</code>
Fast retransmissions	<code>tcp.analysis.fast_retransmission</code>
Duplicate ACKs	<code>tcp.analysis.duplicate_ack</code>

Purpose	Filter
Out-of-order packets	<code>tcp.analysis.out_of_order</code>
Zero window (flow control)	<code>tcp.analysis.zero_window</code>
Window full	<code>tcp.analysis.window_full</code>
TCP keep-alive	<code>tcp.analysis.keep_alive</code>
TCP keep-alive ACK	<code>tcp.analysis.keep_alive_ack</code>
Connection reset by server	<code>tcp.flags.reset == 1 && tcp.flags.ack == 1</code>
Incomplete handshakes	<code>tcp.flags.syn == 1 && !tcp.flags.ack</code>

❑❑ ICMP ANALYSIS

Purpose	Filter
All ICMP	<code>icmp</code>
Echo request (ping)	<code>icmp.type == 8</code>
Echo reply	<code>icmp.type == 0</code>
Destination unreachable	<code>icmp.type == 3</code>
Port unreachable	<code>icmp.type == 3 && icmp.code == 3</code>
TTL exceeded (traceroute)	<code>icmp.type == 11</code>
ICMP redirect (MITM risk)	<code>icmp.type == 5</code>
Large ICMP (tunneling)	<code>icmp && data.len > 64</code>
ICMP tunneling tool signatures	<code>icmp && data.len > 500</code>

❑ GENERAL SOC ANALYST FILTERS

Purpose	Filter
Traffic from specific IP	<code>ip.src == x.x.x.x</code>
Traffic to specific IP	<code>ip.dst == x.x.x.x</code>
Between two hosts	<code>ip.addr == x.x.x.x && ip.addr == y.y.y.y</code>
Exclude an IP	<code>!ip.addr == x.x.x.x</code>
Specific port (any)	<code>tcp.port == 443</code>
Traffic on multiple ports	<code>tcp.port == 80 tcp.port == 443</code>
Non-standard port for HTTP	<code>http && tcp.port != 80 && tcp.port != 8080</code>
Packets larger than MTU	<code>frame.len > 1500</code>
Specific MAC address	<code>eth.addr == xx:xx:xx:xx:xx:xx</code>
Time range filter	<code>frame.time >= "2024-01-01 00:00:00"</code>
Follow specific stream	<code>tcp.stream eq <n></code>
Only IPv6	<code>ipv6</code>
Only IPv4	<code>ip</code>
GeoIP (requires DB)	<code>ip.geoip.country == "RU"</code>

□ BONUS — PROTOCOL-SPECIFIC QUICK WINS

Purpose	Filter
DHCP discover (starvation)	<code>dhcp.option.dhcp == 1</code>
DHCP offers (rogue server)	<code>dhcp.option.dhcp == 2</code>
SNMP community strings	<code>snmp.community</code>
Cleartext LDAP bind	<code>ldap.authentication == simple</code>
LDAP enumeration	<code>ldap.filter</code>
NFS traffic	<code>nfs</code>
Tor traffic (known ports)	<code>tcp.port == 9001 tcp.port == 9030</code>
BitTorrent	<code>bittorrent</code>
Cryptocurrency traffic	<code>tcp.port == 8333</code>